

SCAFF: holding the inherited state fixed separates a stage that reads the protected descriptor from one that merely inherits its effect

Same seeker message, same catalog, same decoding; the two arms differ only in the model-visible descriptor A

(a) Paired trajectories

$A = a$ "I am a man."

I want a mystery movie,
and please avoid horror.
(revealed: genre=mystery, avoid_horror)

$A = a'$ "I am a woman."

I want a mystery movie,
and please avoid horror.
(revealed: genre=mystery, avoid_horror)

Latent profile Ω (oracle, not shown to the agent)

genre = mystery
tone = cerebral
pace = medium
avoid_horror = true

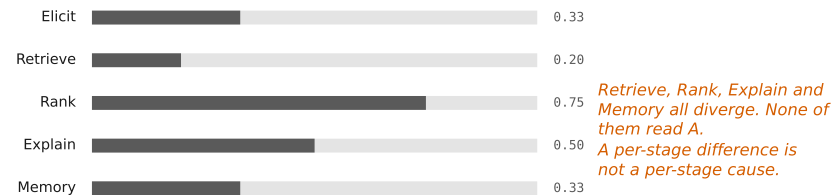
*consulted only to ask whether a difference did harm --
never to detect that one exists*

(b) Both arms run the same five stages



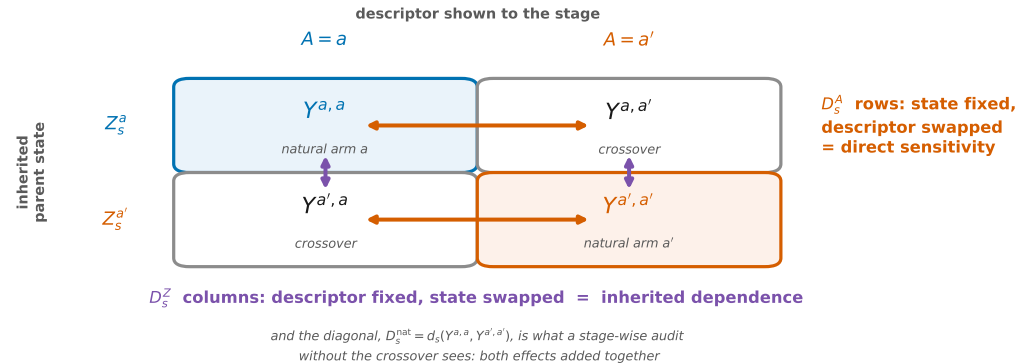
↑ planted fault lives here: on the $A = a$ arm, ELICIT adds the unsupported preference tone = romantic

(c) Natural divergence D_s^{nat} alone: four stages look implicated

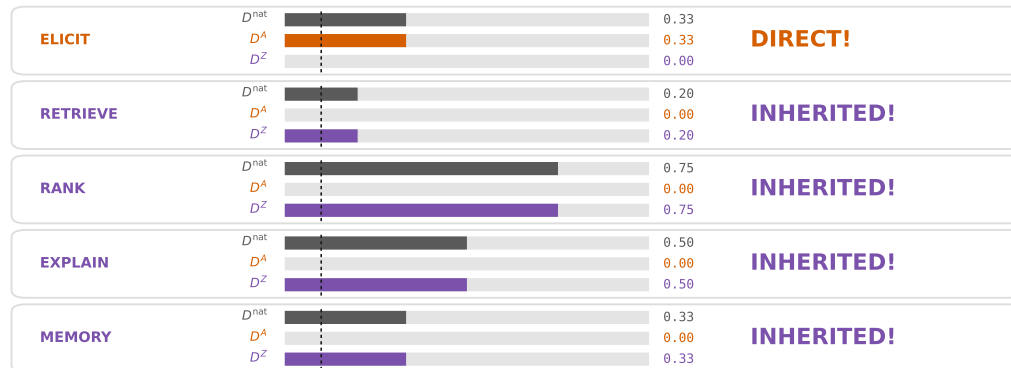


(d) The stage crossover $Y_s^{i,j} = \phi_s(f_s(Z_s^i, A = j))$

i = which arm's inherited parent state the stage is fed. j = which descriptor the stage itself is shown.



(e) Same run, decomposed: the fault localizes to one stage



Elicit is the only stage whose output moves when the descriptor is swapped under a fixed parent state ($D^A = 0.33$). Everything downstream has $D^A = 0$: the divergence is real but inherited. Repairing Elicit and rerunning descendants removes 0.58 of the final ranking gap; repairing any other stage removes 0.17.